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This document outlines and breaks down the scoring algorithm used by Salud.

1. Overview

Algorithm Name: Salud Health Score

Purpose:

This algorithm is used to assess the healthiness of an individual food item by providing a score from 0-1. It takes into account user preferences and is designed to be tunable.

Scope:

This algorithm is used after a food item is scanned and its raw data is stored.

Constraints:

1. Does not know the exact quantity of each ingredient
2. Ingredient quality is dependent on a predetermined score (can change over time) and some ingredients may not be accounted for

2. High-Level Description

Conceptual Summary:

This algorithm outputs a score that is made up of the 4 weighted scores based on the following areas:

1. Energy Ratio
2. Macro Ratio
3. Ingredient Quality
4. Nutrient Density

Each of these sub scores are weighted depending on their impact on health. These weights can be tweaked over time to improve the accuracy of the algorithm.

1. Energy Ratio Score:

This sub score is based on the caloric density of a given food item. This score is then weighted based on the user's goals. (ex. A user who is trying to lose weight's score will be more heavily impacted by a calorically dense food)

2. Macro Ratio Score

This sub score is based on how well a given food item's macro ratio matches the user's macro goals.

3. Ingredient Quality Score:

This sub score is based on the following factors: ingredient risk factor & ingredient quantity. Ingredient risk factor is a predetermined score based on the current research. Ingredient quantity is the position in the ingredient list. The exact ingredient quantity can not be determined without reverse engineering a food which is both unfeasible and unreliable.

4. Nutrient Density Score:

This sub score reflects how nutritious a food is. This score can be tuned depending on how much of an emphasis on nutrient density is desired.

3. Algorithm Steps

This algorithm is broken into 5 parts

1. Compute Energy Ratio Sub Score
2. Compute Macro Ratio Sub Score
3. Compute Ingredient Quality Sub Score
4. Compute Nutritional Density Sub Score
5. Compute Overall Health Score

3.1 Energy Ratio Sub Score

Type	Name	Description	Format/Example
Output	Caloric Density	Calories per gram of food	d_c
Output	Energy Ratio Sub Score	Decimal from 0-1 that represents a good or bad energy score	s_E
Input	Calories	Number of calories per 100g of food	c
Input	Energy Score Constant	Decimal from 0-1 that controls how steep score fall off is (Note: this weight can vary depending on a user's goals)	k_E

$$d_c = \frac{c}{100}$$

$$s_E = \frac{1}{1+k_E d_c}$$

3.2 Macro Ratio Sub Score

Type	Name	Description	Format/Example
Output	Macro distance	Total weighted difference between food	D

		macros and goal macros	
Output	Macro score	Final normalized score (higher = better match)	s_M
Input	Maximum distance	Maximum possible value of D used for normalization	$D_{\max}$
Input	Macro index	Index representing each macro (carbs, protein, fat)	$i \in \{c, p, f\}$
Input	Food macro proportion	Proportion of a given macro in the food (must sum to 1)	f_i
Input	Goal macro proportion	Desired proportion of a given macro (must sum to 1)	g_i
Input	Shortage amount	Amount the food is below the goal for a macro	$\max(0, g_i - f_i)$
Input	Surplus amount	Amount the food exceeds the goal for a macro	$\max(0, f_i - g_i)$
Input	Shortage weight	Weight controlling how strongly shortages are penalized	w_i^{short}
Input	Surplus weight	Weight controlling how strongly excess is penalized	$w_i^{surplus}$
Input	Shortage exponent	Controls how aggressively large shortages are punished (nonlinearity)	α

$$D = \sum_i (w_i^{shortage} \cdot \max(0, g_i - f_i)^\alpha + w_i^{surplus} \cdot \max(0, f_i - g_i))$$

$$s_M = 1 - \frac{D}{D_{\max}}$$

3.3 Ingredient Quality Sub Score

Type	Name	Description	Format/Example
Output	Normalized Risk Score	A conversion of the initial Risk Score normalized to a decimal from 0-1	$\hat{r}_i$
Output	Ingredient Weight	A weight assigned to an ingredient based on its position in the ingredient list	w_i

Output	Ingredient Quality Sub Score	A decimal from 0-1 that represents a good or bad ingredient quality score	s_Q
Input	Risk Score	A score from 0-10 that assess how risky the consumption of that ingredient is (Note: this score is pulled from a database)	r_i
Input	Ingredient Quality Constant	A constant that can be used to tune the score. It is used in the calculation of w_i .	k_Q

$$\hat{r}_i = \frac{r_i}{10}$$

$$w_i = k_Q^{i-1}, \text{ where } 0 < k_Q < 1$$

$$s_Q = 1.0 - \frac{\sum_{i=1}^n w_i \hat{r}_i}{\sum_{i=1}^n w_i}$$

3.4 Nutritional Density Sub Score

Type	Name	Description	Format/Example
Output	Nutrient Score	Overall normalized nutrition score (0–100 scale)	s_N
Input	Nutrient Index	Iterates over each nutrient in the calculation	i
Input	Nutrient Amount	Amount of nutrient i in the food (e.g., grams, mg)	N_i
Input	Daily Value	Recommended daily intake for nutrient i	DV_i
Input	Nutrient Ratio	Percent of daily value provided by the food	$\frac{N_i}{DV_i}$
Input	Capped Nutrient Contribution	Nutrient contribution capped at 100% DV to prevent dominance	$\min(\frac{N_i}{DV_i}, 1)$
Input	Total Nutrient Count	Total number of nutrients being evaluated	N
Input	Sum of Nutrient Contributions	Total of all capped nutrient contributions (density component)	$\sum_i \min(\frac{N_i}{DV_i}, 1)$

Input	Nutrients Above Threshold	Count of nutrients that exceed 5% DV	$\#(N_i/DV_i > 0.05)$
Input	Variety Component	Fraction of nutrients that meaningfully contribute (>5% DV)	$\frac{\#(N_i/DV_i > 0.05)}{N}$

$$s_N = 100 \cdot \left(\frac{\sum_i \min(N_i/DV_i, 1)}{N} \right) \cdot \left(\frac{\#(N_i/DV_i > 0.05)}{N} \right)$$

3.5 Overall Health Score

Type	Name	Description	Format/Example
Output	Overall Health Score	A decimal from 0-1 that represents how healthy a food item is. It comprises the above 4 sub scores.	s_H
Input	Energy Ratio Sub Score	Decimal from 0-1 that represents a good or bad energy score	s_E
Input	Macro score	Final normalized score (higher = better match)	s_M
Input	Ingredient Quality Sub Score	A decimal from 0-1 that represents a good or bad ingredient quality score	s_Q
Input	Nutrient Score	Overall normalized nutrition score (0–100 scale)	s_N
Input	Energy Score Weight	A decimal from 0-1 that represents how much emphasis on energy the final score represents	w_E
Input	Macro Score Weight	A decimal from 0-1 that represents how much emphasis on macro ratio matching the final score represents	w_M
Input	Ingredient Quality Score Weight	A decimal from 0-1 that represents how much emphasis on ingredient quality the final score represents	w_Q
Input	Nutritional Density Score Weight	A decimal from 0-1 that represents how much emphasis on nutritional density the final score represents	w_N

$$s_H = s_E^{w_E} \cdot s_M^{w_M} \cdot s_Q^{w_Q} \cdot s_N^{w_N}$$